FINAL AUTOMATION FLOW, IMPACT REPORT & WALKTHROUGH

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Objectives: Bring together the entire automation project into one clear and complete solution. It explains how the system works from start to finish, including the triggers, actions, and decision paths that allow it to run automatically. It also evaluates the real impact of the automation by comparing the previous manual process with the new automated workflow, highlighting improvements in time, efficiency, and accuracy, while acknowledging any limitations. Finally, it provides a simple, step-by-step walkthrough supported by visuals or a demo to ensure that both technical and non-technical audiences can easily understand how the automation adds value to the business.

I. INTRODUCTION

Many organizations still rely on manual processes to collect and manage data, especially when information comes from different sources. In this case, sales orders were received through phone calls, emails, and online forms, and staff had to manually read each order, enter the details into spreadsheets, verify the information, and consolidate everything for reporting. Although this process worked, it required significant time and effort, increased the risk of errors, and became harder to manage as order volume grew. To improve efficiency and accuracy, a multi-channel automation system was developed to standardize order intake, automatically validate and process transactions, centralize data into a single master sheet, and enable real-time dashboard reporting. This final deliverable presents the completed automation workflow, evaluates its impact compared to the previous manual process, and demonstrates how the solution enhances operational efficiency, data reliability, and decision-making visibility.

II. FINAL WORKFLOW DOCUMENTATION

2.1 Automation Logic

The automation system handles sales orders from Email, Phone, and Google Forms by automatically validating and standardizing the data, consolidating it into a centralized master dataset, and enabling automated reporting. The logic is designed to ensure accuracy, prevent duplication, and keep all order sources synchronized in real time. To make this possible, Google Apps Script serves as the core engine that runs the automation in the background. Through built-in time-driven and event-driven triggers, the system automatically executes tasks such as extracting order details, checking required fields, calculating totals, generating structured Order IDs, and updating the centralized sheet. These scripts run continuously without manual intervention, ensuring that new or edited records are processed instantly and that reports always reflect the latest data. By integrating Apps Script into the workflow, the automation becomes a fully functioning, end-to-end system that is reliable, scalable, and efficient.

2.1.1 Trigger Layer (Event Activation)

The system begins with automated triggers that detect new or edited data without manual intervention:

- **Email Orders** – Time-driven trigger (runs every minute) to scan unread Gmail messages with a defined subject format.
- **Phone Orders** – Event-driven triggers (`onEdit` and `onChange`) to detect new entries, edits, or row deletions in salesperson sheets.
- **Form Orders** – Time-driven trigger (runs every minute) to process new Google Form submissions in batch mode.
- **Consolidation Engine** – Time-driven trigger (runs every minute) to synchronize all validated records into the centralized dataset.
- **Reporting Layer** – Scheduled trigger in Looker Studio to automatically generate and distribute dashboard reports.

2.1.2 Channel-Level Processing Logic

Once activated, each channel follows a structured validation and processing sequence.

a. Data Extraction

The system extracts required fields such as:

- Order Date
- Customer Name
- Contact Details
- Product Specifications
- Quantity
- Location

Each field is mapped into a standardized format to ensure consistency across channels.

b. Validation Engine

Before any record is accepted, the automation enforces strict validation rules:

- Required fields must not be blank
- Quantity must be greater than zero
- Product must exist in the PRODUCT_MASTER sheet
- Phone numbers must follow strict formatting rules
- Dates must pass safe parsing validation
- Currency values are sanitized before calculation

If validation fails, the system stops processing that record and prevents corrupted data from entering the system.

c. Financial Computation Logic

For valid records:

- Unit Price is retrieved from PRODUCT_MASTER
- Currency formatting is sanitized
- Order Amount is computed as:
- Order Amount = Quantity × Unit Price
- Calculations only execute when both inputs are valid numeric values

This eliminates common errors such as #NUM! or malformed currency computations.

d. Automated Salesperson Assignment

The system automatically assigns the salesperson based on predefined mapping rules before the Order ID is generated. While the location is defined directly by the customer at the time of order, the automation uses that location input to determine the corresponding assigned salesperson according to a predefined territory or synthetic company configuration.

This ensures that each transaction is consistently attributed to the correct sales representative without requiring manual selection. The automated assignment reduces errors, standardizes performance tracking, and maintains reporting integrity across all channels.

e. Structured Order ID Generation

Each channel generates a unique structured Order ID:

- EO – Email Orders
- PO – Phone Orders
- FO – Form Orders

The system scans existing IDs globally to maintain continuous sequencing and prevent duplication. Order IDs are generated only once per valid transaction to prevent overwriting.

After the salesperson is auto-assigned and all required fields are validated, the system generates a unique structured Order ID. The ID format is constructed using:
Location Abbreviation + Order Date + Order Source (EO/PO/FO) + Continuous Series Number

The system scans existing Order IDs globally to maintain continuous sequencing and prevent duplication. Order IDs are generated only once per valid transaction to preserve data integrity and prevent overwriting.

f. Timestamp Control

A completion timestamp is created only once, when the Order ID is first generated, to preserve the original record of when the transaction was completed. This timestamp remains unchanged even if the order details are edited later, ensuring that the original completion time is protected for reporting and audit purposes. However, any valid updates made to the order are still automatically reflected in the consolidated sheet through the Live Mirror process. This allows the system to stay up-to-date with the latest information while maintaining the integrity of the original transaction record.

2.1.3 Live Mirror Consolidation Logic

The consolidation engine performs a full rebuild synchronization process:

a. Extract validated records from:
- Email_Orders
- Phone MASTERFILE
- ORDER_FORM (Responses)

b. Remove headers
c. Validate completion timestamps
d. Normalize all date formats
e. Merge all records in memory
f. Sort by completion timestamp (latest first)
g. Clear Centralized_Order sheet
h. Rewrite the entire dataset in batch mode

This "Live Mirror" model ensures:
- Deleted records are removed from the master
- No duplicate records persist
- Edits are reflected accurately
- Chronological order is maintained

The Centralized_Order sheet becomes the single source of truth.

2.1.4 Reporting and Distribution Logic

The centralized dataset connects directly to Looker Studio. Once the master sheet refreshes:
- Dashboard metrics update automatically
- Sales trends, category performance, and regional analysis refresh in real time
- A scheduled email trigger generates a snapshot report
- The report is automatically distributed to stakeholders

This eliminates the need for manual exporting and report emailing.

2.2 Summary of Triggers, Actions and Decision Paths

Category	Component	Description	System Behavior
Trigger	Email Orders	Time-driven trigger (runs every minute)	Scans unread Gmail orders and initiates processing
	Phone Orders	Event-driven (onEdit, onChange)	Activates when a row is added, edited, or deleted
	Form Orders	Time-driven trigger	Processes new form submissions in batch mode
	Consolidation Engine	Time-driven trigger	Synchronizes validated records into master sheet
	Reporting	Scheduled (Looker Studio)	Automatically generates and distributes reports

Category	Action	Description	Purpose
Action	Field Extraction	Parse and map incoming order data	Standardize structure across channels
	Validation Engine	Check required fields, format, product match	Prevent incomplete or invalid records
	Auto Salesperson Assignment	Assign salesperson based on customer location	Ensure consistent sales attribution
	Financial Computation	Calculate Order Amount (Qty × Unit Price)	Eliminate manual calculation errors
	Structured Order ID Generation	Create ID (Location + Date + Source + Series No.)	Ensure uniqueness and traceability
	Timestamp Creation	Generate completion timestamp once	Preserve transaction integrity
	Validated Sheet Append	Insert fully validated transactions	Prevent corrupt entries
	Live Mirror Consolidation	Merge, normalize, sort, rebuild dataset	Maintain single source of truth
	Dashboard Refresh	Pull updated centralized data	Enable real-time visibility
	Report Distribution	Email scheduled dashboard snapshot	Eliminate manual reporting

Category	Decision Point	If Condition Fails	If Condition Passes
Decision Path	Required Fields Check	Reject / Skip record	Proceed to product validation
	Product Match Check	Reject record	Proceed to financial computation
	Financial Validation	Do not compute / Stop insertion	Proceed to Order ID generation
	Duplicate ID Check	Generate next sequence	Assign unique ID
	Consolidation Validation	Exclude from master	Include in Centralized_Order

2.3 End-to-end Workflow Summary

Customer Order → Trigger Activation → Field Extraction → Validation → Financial Computation → Structured ID Generation → Timestamp Assignment → Validated Sheet Append → Live Mirror Consolidation → Centralized_Order Update → Dashboard Refresh → Automated Report Distribution

Integrated End-to-End Automation Architecture

EMAIL ORDERS (Gmail)

PHONE ORDERS (Sales Sheets)

FORM ORDERS (Google Form)

Triggered Automatically via Apps Script

Channel-Level Processing Engines

- Feld Extraction
- **Validation Engine** (Required Fields / Format Checks)
- Auto Salesperson Assignment
- Financial Computation
- **Structured Order ID Generation** (Loc + Date + Source + No.)
- Timestamp Creation

Validated Channel Outputs

• Only validated transactions proceed •

- Email_Orders
- Phone MASTERFILE
- **ORDER_FORM** (Responses)

Live Mirror Consolidation

- Extract from all validated sheets
- Normalize dates
- Sort by completion timestamp
- Full dataset rebuild

Centralized_Order *(Single Source of Truth)*

- Single Source of Truth
- Reporting-Ready Dataset

Visualization & Report Distribution

- Looker Studio Dashboard
- Scheduled Email Report Distribution

III. IMPACT ANALYSIS REPORT

3.1 Comparison of Manual vs Automated Process

Process Area	Manual Workflow	Automated Workflow	Impact
Order Intake	Orders received via phone, email, or form and manually read by staff	Apps Script triggers automatically detect and process incoming orders	Eliminates manual monitoring and reduces processing delay
Data Entry	Staff manually encode data into spreadsheets	Automated field extraction and mapping into structured sheets	Reduces encoding time and minimizes human error
Validation	Manual checking of required fields and formats	Automated validation engine enforces required fields and format rules	Prevents incomplete or invalid entries before insertion
Product Matching	Manual cross-check with product list	Automatic lookup from PRODUCT_MASTER	Ensures consistency and eliminates mismatch errors
Salesperson Assignment	Manually selected by staff	Auto-assigned based on predefined location mapping	Standardizes sales attribution and removes selection errors
Financial Computation	Manual calculation (Qty × Unit Price)	Automated computation with currency sanitization	Eliminates calculation errors and #NUM! issues
Order ID Generation	Manually created or based on row count	Structured ID generated using location, date, source, and sequence logic	Prevents duplicate IDs and ensures traceability
Timestamp Recording	Manually entered or inconsistent	System-generated and preserved timestamp	Maintains accurate transaction history
Consolidation	Manual merging of multiple sheets	Live Mirror consolidation with full rebuild logic	Removes duplication, ensures synchronization
Reporting	Manual dashboard refresh and email distribution	Real-time dashboard update and scheduled automated reports	Eliminates manual reporting effort and delays
Error Handling	Errors discovered after reporting	Errors prevented at validation stage	Reduces operational risk
Scalability	Performance decreases as volume increases	Trigger-based automation handles increasing data volume	Supports growth without proportional manpower increase

3.2 Time, Effort, and Efficiency Improvements

3.2.1 Time Improvements

Activity	Manual Process	Automated Process	Improvement
Order Encoding	3–5 minutes per order (reading, typing, checking)	Instant automated extraction	Approximately 60–70% faster processing
Validation Checking	Manual review per entry	Real-time script validation	Immediate error detection
Financial Computation	Manual multiplication and correction	Automated computation	Eliminates recalculation time
Consolidation	Manual merging of multiple sheets (30–60 mins daily)	Live Mirror automatic rebuild	100% elimination of manual consolidation
Report Preparation	Manual dashboard refresh and PDF emailing	Scheduled automatic distribution	Near-zero manual reporting time

Summary:
The system transitions from manual batch handling to near real-time automation, which significantly reduces the turnaround time from order intake to executive reporting.

3.2.2 Effort Improvements

Activity	Manual Effort	Automated Effort	Improvement
Data Monitoring	Staff monitor inbox, sheets, and forms	System monitors via triggers	Removes continuous manual oversight
Error Correction	Manual identification and fixing	Validation prevents errors before entry	Reduces rework
Salesperson Assignment	Manually selected	Automatically assigned	Removes selection workload
ID Management	Manually tracked sequence	System-generated structured ID	Prevents duplication and manual tracking
Reporting Distribution	Manual download and email	Automated scheduled dispatch	Eliminates repetitive reporting task

Summary:
Staff effort shifts from repetitive administrative work to higher-value tasks such as analysis and decision-making.

3.2.3 Efficiency Improvements

Efficiency Dimension	Manual Process	Automated Process	Improvement
Processing Speed	Orders processed after manual review	Orders processed instantly via triggers	Real-time transaction handling
Data Accuracy	Prone to human encoding and calculation errors	Validation engine prevents invalid entries	Significant reduction in data errors
Financial Computation	Manual multiplication and correction	Automatic sanitized computation	Eliminates recalculation and correction time
Order ID Management	Manually tracked sequence, risk of duplication	Structured global ID generation	Prevents duplicates and sequence resets
Salesperson Assignment	Manual selection	Auto-assigned based on predefined mapping	Standardized and error-free attribution
Data Consolidation	Manual merging across sheets	Live Mirror full dataset rebuild	100% elimination of manual consolidation
Reporting Cycle	Manual dashboard refresh and emailing	Automated dashboard refresh and scheduled distribution	Continuous executive visibility
Error Detection	Errors discovered after entry	Errors blocked during validation stage	Prevents downstream correction effort
Operational Dependency	High reliance on human oversight	System-driven trigger execution	Reduced administrative workload
Scalability	Processing slows as order volume increases	Trigger-based automation handles growth	Improved scalability without added manpower

Summary:
The automation increases operational efficiency by shifting the workflow from manual, sequential processing to automated, trigger-based execution. Tasks that previously required repetitive effort, monitoring, and correction are now system-controlled, resulting in faster processing time, improved data integrity, reduced manual intervention, and scalable operations.

3.3. Observed Benefits and Limitations

Process	Benefits	Limitations
Order Processing	Near real-time processing through trigger-based automation	Dependent on Apps Script trigger execution limits
Data Accuracy & Validation	Automated validation prevents incomplete, invalid, and duplicate entries	Validation logs require monitoring for visibility
Financial Computation	Eliminates manual calculation errors and currency formatting issues	Relies on correct PRODUCT_MASTER data maintenance

Salesperson Assignment	Consistent, automated sales attribution based on predefined mapping	Requires updated mapping rules if territories or location change
Order ID Management	Structured global sequencing prevents duplication and maintains traceability	May require counter optimization at very high transaction volumes
Data Consolidation	Live Mirror rebuild ensures single source of truth and accurate synchronization	Full rebuild logic may impact performance with very large datasets
Reporting & Visibility	Real-time dashboard updates and automated report distribution	Dependent on Google Workspace and Looker Studio environment
Scalability	Handles increasing order volume without proportional manpower increase	Further optimization may be needed for enterprise-scale expansion
Audit & Traceability	Preserved timestamps maintain transaction integrity	No advanced audit log dashboard yet implemented
Operational Efficiency	Reduces manual workload and administrative repetition	Initial setup requires technical configuration knowledge

Summary:
Overall, the automation transforms a manual and error-prone order process into a structured, efficient, and reliable system by enabling real-time processing, automated validation, standardized ID generation, salesperson assignment, centralized consolidation, and automated reporting. It significantly reduces manual workload, improves data accuracy, and enhances decision-making visibility through a single source of truth. While the system depends on the Google Workspace environment and may require optimization as transaction volume scales, its benefits in efficiency, control, and scalability clearly outweigh its limitations, providing a strong operational foundation for streamlined sales management.

IV. AUTOMATION WALKTHROUGH

4.1 Over-the-Phone Order

Demo Link:
https://drive.google.com/file/d/1K_PM7Og5dHoncFFXrEd5yZJEflfPat54/view?usp=sharing
Scripts Link:
https://drive.google.com/file/d/17nbJjIq6mmJeABSRVQLbJcRU7U61-32d/view?usp=sharing
Data Storage Link:
https://docs.google.com/spreadsheets/d/1-2PRBt08G8i1Bsk9-Ktb1mZLDKHVPLw7K3DTTsKF5e8/edit?usp=sharing
Overview:
The Phone Order Automation System transforms manual phone order encoding into a structured, automated, and centrally controlled workflow using Google Sheets integrated with Apps Script. It functions as a lightweight order management system that standardizes product selection, automates financial calculations, generates structured order IDs, and consolidates validated orders into a centralized data warehouse.

4.2 Over-the-Form Order

Demo Link:
https://drive.google.com/file/d/10_E2f9xLC7BECn5ZmCjX-wEFN8eX2unH/view?usp=sharing

Scripts Link:
https://drive.google.com/file/d/1haJVlR5XH1KluWPzxahW3-5Wqco04NRX/view?usp=sharing

Order Form Link: https://forms.gle/ufsADBMkLMXe5VZZ8

Data Storage Link:
https://docs.google.com/spreadsheets/d/1CJrkLYMZbdFQ6i9VH040ZwcON44phgrneFMSkjAU6so/edit?usp=sharing

Overview:

The Form Order Automation System converts Google Form submissions into a structured, fully automated order processing workflow using Google Sheets integrated with Apps Script. It operates as a lightweight order management system that automatically extracts validated customer inputs, performs intelligent product lookups from a centralized product master database, calculates financial values, generates standardized order IDs, assigns territory-based sales representatives, and synchronizes validated transactions into a consolidated data repository. This architecture transforms raw form responses into production-ready order records with built-in validation, reconciliation, and traceability controls.

4.3 Over-the-Email Order

Demo Link:
https://drive.google.com/file/d/157yITNvGc0JzOJTsrq6KH309LN7Cc-LP/view?usp=sharing

Scripts Link:
https://drive.google.com/file/d/1u5liK8x6jkmqY761kbxPdFkMFbp1RgEt/view?usp=sharing

Data Storage Link:
https://docs.google.com/spreadsheets/d/1XA2P3e_QZzO5Qkwsie9kxmHE9St-zVcTyvMsO89p2XA/edit?usp=sharing

Overview:

The Email Order Automation System converts structured customer email orders into a validated, automated, and centrally managed workflow using Google Sheets integrated with Google Apps Script and Gmail. It functions as a lightweight email-driven order management engine that parses incoming emails, validates product specifications against a master database, enforces standardized formatting rules, generates continuous structured Order IDs, performs automatic financial calculations, and logs clean, verified transactions into a centralized reporting sheet. The system operates on a timed trigger, scanning unread emails with a defined subject format and transforming raw email content into structured transactional records without manual encoding.

4.4 Centralized Order Data

Demo Link:
https://drive.google.com/file/d/1n7OX-U3uvRdI4nuLSAvTqRCu9ev74EAP/view?usp=sharing

Scripts Link:
https://drive.google.com/file/d/1J4N_MQ2dVHA0T4GVCkNs0uG14XS0NWpS/view?usp=sharing

Data Storage Link:
https://docs.google.com/spreadsheets/d/16DvAezJeAJdiQoUICx9dEjJHhn3XfWIiTSWud29jvHg/edit?usp=sharing

Overview:
The Centralized Order Live Mirror System transforms fragmented multi-channel order intake (Email, Phone, and Google Forms) into a synchronized, automated, and centrally governed data workflow using Google Sheets integrated with Apps Script. It functions as a real-time consolidation engine that standardizes order structures across channels, enforces data validation, performs forced date normalization, and dynamically rebuilds a master reporting sheet that mirrors all source transactions. The system ensures that the MASTERFILE remains an exact reflection of all operational order sources, automatically capturing edits, updates, and deletions while maintaining accurate chronological sorting based on Completion Order for reporting and dashboard integration.

4.5 Report Distribution Data

Demo Link:
https://drive.google.com/file/d/14bq12sg9TiGdBTvgiq2k0uTByN77v0OU/view?usp=sharing

Report Link:
https://lookerstudio.google.com/reporting/ee01af17-7ce5-454a-9214-09da4ec3c2f7/page/page_12345

Synthetic Company Names Link:
https://docs.google.com/spreadsheets/d/1u5Az7dhrCSl3ctdR5pS0_XXtJeQEPALksPt46l1ySYE/edit?usp=sharing

Synthetic Salesperson Assignment Link:
https://docs.google.com/spreadsheets/d/1cM-jpjB41xp-fuHaYiwe-czUh1j-JD1n8J1GkIoGbiM/edit?usp=sharing

Overview:
The Automated Report Distribution System transforms static dashboard viewing into a structured, scheduled, and centrally managed reporting workflow using Google Sheets as the data source and Looker Studio as the visualization layer. Integrated with scheduled email delivery, it enables automated generation and distribution of PDF reports to stakeholders. The system ensures standardized reporting metrics, real-time data synchronization, and consistent executive-level visibility without manual intervention.

4.6 Workflow Explanation Summary

Section	What It Does (Simple Explanation)
Over-the-Phone Order	When a staff member encodes a phone order, the system automatically validates the details, calculates the total amount, assigns the correct salesperson, and generates a structured Order ID. The validated order is then saved into a centralized sheet without requiring manual consolidation. This removes calculation errors and standardizes phone-based transactions.
Over-the-Form Order	When a customer submits a Google Form, the system automatically reads the response, verifies product details, computes the total price, assigns the correct salesperson, and creates a standardized Order ID. The processed order is stored in a structured dataset ready for reporting. This transforms raw form submissions into complete, production-ready order records.
Over-the-Email Order	When a customer sends an order via email, the system scans unread messages, extracts order details, validates product information, calculates totals, and generates a structured Order ID. It converts unstructured email content into clean, organized transaction data automatically. This eliminates the need for manual email encoding.
Centralized Order Data	All validated orders from Phone, Form, and Email are automatically merged into one master sheet. The system ensures the data is synchronized, updated in real time, and accurately sorted for reporting. This sheet acts as the single source of truth for the entire operation.
Report Distribution Data	The centralized data feeds into a dashboard that updates automatically. Reports are generated and sent to stakeholders on a scheduled basis without manual downloading or emailing. This ensures consistent, real-time executive visibility.

V. CONCLUSION

This project demonstrates how a manual and time-consuming sales order process can be transformed into a streamlined and reliable automated system. By connecting multiple order channels into one structured workflow, the solution reduces errors, saves time, and provides real-time visibility for better decision-making. Instead of relying heavily on manual encoding, checking, and reporting, the system now handles these tasks automatically and consistently. Overall, the automation not only improves efficiency and accuracy but also creates a scalable foundation that supports smoother operations and long-term business growth.

VI. APPENDIX

Presentation Slide Link: *click or select to access*
Presentation Recording Link: *click or select to access*